All 16 refineries, 5,108 million gal/year — Current + lag-one stock: combined response

Inverse-square Exponential Lowest-cost only

0–25 miles

25–50 miles

50–75 miles

75–100 miles

100–125 miles

125–150 miles

Road
Straight line
Mode cost
Rail cost ×0.5
Rail cost ×2

All-refinery share contribution (pp)

Actual 2024 eligible capacities and locations. Separate rings; 2024 non-water area weights; matched 2016–2025 sample.
95% crusher-clustered coefficient-only intervals. Model-implied contribution versus zero exposure; no established national causal counterfactual.